JAVIER A. CUEVAS CHABRIER

(352) 693-0477 | javy.acc@gmail.com | javy-scratchspace.github.io/javy0119_portfolio.io | Secret Clearance

Education

University of Central Florida (UCF) Expected Graduation: May 2029

B.S. Mechanical Engineering, B.S. Computer Science. GPA: 3.42

Research and Industry Experience

Propulsion and Energy Research Lab (PERL) | **Propulsion Engineering Research Assistant** *Jan 2025 – Present*

- Improved lab calculation efficiency by 50% and reduced errors by developing a Python-based calculator using Cantera to solve reactant requirements for target chamber pressure, velocity, and momentum flux.
- Evaluated Oblique Detonation Wave Engine Vehicle component performance at Mach 10 using ANSYS Fluent and analyzed fuel and oxidizer properties from Axial Engine test data.
- Improved calculator reproducibility and engineering knowledge transfer by creating comprehensive process documentation and explaining the underlying calculation methodology to other engineers.

Lockheed Martin | **Systems and Software Engineer** *Sep 2025 – Present*

- Improved simulation visualization and accelerated model-generation workflows by automating model creation with AI-driven source-code parsing and Python implementation.
- Reduced plugin errors through a feedback-based improvement system and cut project completion times from days to minutes. This improved the performance and reliability of automated engineering workflows and informed adopters about the plugins' applications.

Partnerships for Research and Education in Materials | **Physics Research Assistant** *Sep 2024 – Oct 2025*

- Built Python automation tools for lab data, which reduced data analysis timeframes by 75%.
- Operated AFM, CVD, and a thermal evaporator. Built a transfer stage using heaters and a PID controller.
- Studied the growth kinetics and transfer mechanics of monolayer MoS_2.

Leadership and Organizations

Society of Hispanic Professional Engineers (SHPE) | **Competitions Director**

- Managed a group of 30+ students to effectively prepare and develop solutions to problem prompts by guiding them through the Engineering design process. Demonstrated proper documentation procedures.

UCF Knights Experimental Rocketry (KXR) – **Launch and Test Infrastructure Team**

- Supported software and electronics development for the Launch, Test, and Infrastructure team. Assisted with sensor development, hardware/software testing, and system bring-up.

UCF HSI Battle of the Brains | **Team Captain**

- Led technical solution design and implementation for the 2025 season and managed a 7-person team in 2026, ensuring scope alignment and on-time delivery with built-in review time.

Projects and Competitions

- **Solid Rocket Motor Project:** Designed and fabricated solid propellant rocket motors. Ran static-fire tests and used data analysis to optimize thrust output and structural integrity.
- **Electric Generator Project:** Designed and built an electric generator utilizing copper coils, magnetic assemblies, and custom 3D-printed parts derived from personal CAD designs and previous research. The resulting generator produced 0.5 A; future development plans include increasing wattage output.

Skills

Hardware/Software: soldering, 3D printing, laser cutting, Python, C++, MATLAB, power tools